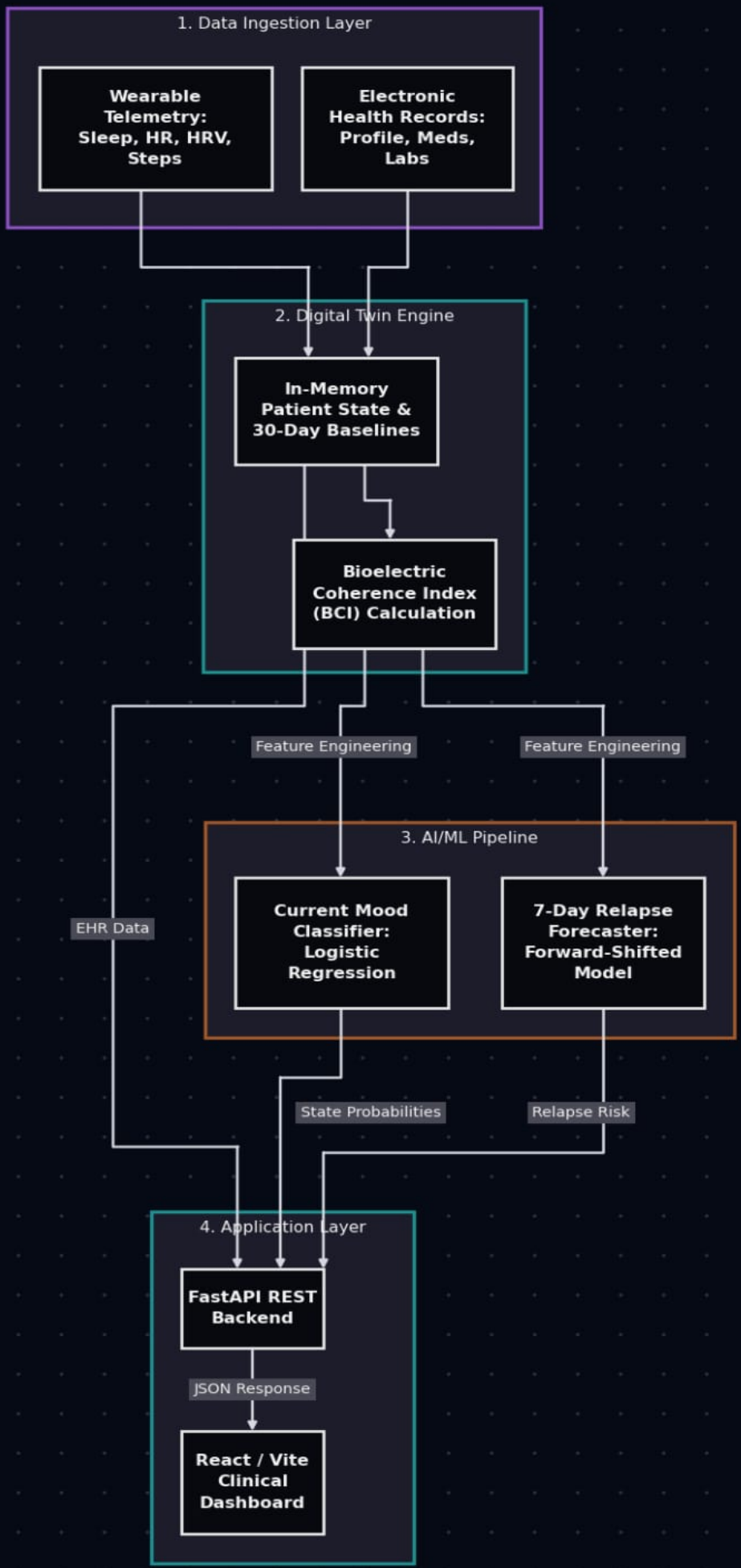

1. Data Ingestion Layer
Wearable Telemetry: Sleep, HR, HRV, Steps
Electronic Health Records: Profile, Meds, Labs
2. Digital Twin Engine
In-Memory Patient State & 30-Day Baselines
Bioelectric Coherence Index (BCI) Calculation
Feature Engineering
Feature Engineering
3. AI/ML Pipeline
EHR Data
Current Mood Classifier: Logistic Regression
7-Day Relapse Forecaster: Forward-Shifted Model
State Probabilities
Relapse Risk
4. Application Layer
FastAPI REST Backend
JSON Response
React / Vite Clinical Dashboard